

HONEYWELL EMBEDDED ENGINEERING TEAMS — PA · IA · BA

AI Champions Programme · The Capstone

Module 12

Honeywell Embedded Engineering Capstone and Adoption Roadmap

Eleven modules, one integrated use case: spec through implementation, test evidence, MCP workflow, PR quality, monitoring, ROI — and a 90-day plan to make it real.



DURATION

2 Hours · Module 12 of 12



FORMAT

Team Capstone Presentations



DELIVERED FOR

Honeywell Embedded Engineering

How We'll Spend These Two Hours

Six connected moves — from the full journey recap to your team standing up to present.



01

The Whole Journey, One Page

10 min

A compact recap of all eleven modules — what each one built, in one line.



02

What the Capstone Contains

10 min

The ten components every team's blueprint must include, and why.



03

The Evaluation Rubric

10 min

What “excellent” looks like, so nobody's guessing what's being scored.



04

The 90-Day Adoption Roadmap

10 min

Turning a capstone project into a realistic rollout plan.



05

Build & Rehearse

60 min

Assemble evidence, build the blueprint, and rehearse the presentation.



06

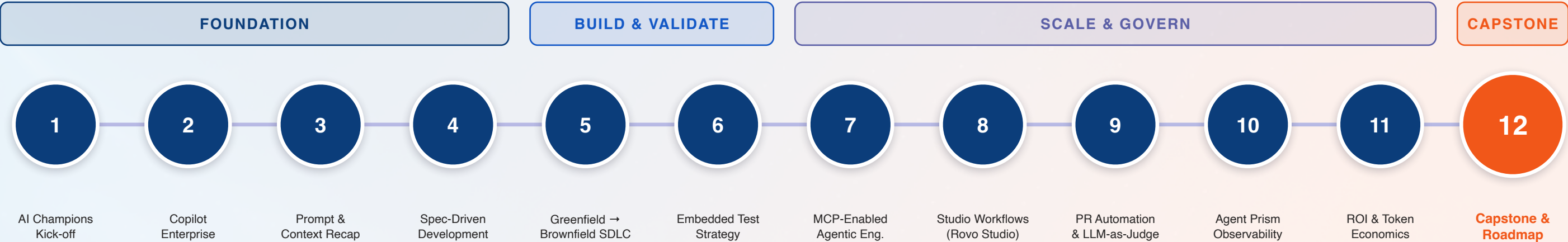
Team Presentations

20 min

Present, get evaluated against the rubric, and close the programme.

Module 12 in the 12-Module Journey

The last node — where everything built since Module 01 gets demonstrated as one working whole.



Why this matters: There's no new engineering content today — every capability your capstone demonstrates was built in Modules 01–11. This module is where it gets proven, in public, against a real use case.

The Whole Journey, One Page

Eleven modules, each in one line — what you actually walk away able to do.

01

AI Champions Kick-off

Named the risks, set the success metrics

02

Copilot Enterprise

Mapped Copilot's real strengths & limits

03

Prompt & Context Recap

Scoped context on purpose, not by accident

04

Spec-Driven Development

Wrote specs a test can actually prove

05

Greenfield → Brownfield SDLC

Built new, evolved existing, safely

06

Embedded Test Strategy

Proved a planted defect gets caught

07

MCP-Enabled Agentic Eng.

Connected agents to real tools, governed

08

Studio Workflows (Rovo Studio)

Gave business roles a governed way in

09

PR Automation & LLM-as-Judge

Turned good instincts into an automatic gate

10

Agent Prism Observability

Made agent activity visible, not assumed

11

ROI & Token Economics

Turned engineering discipline into a business case

12

Capstone & Roadmap

Proving it all together — today

From Day One to Now: Built From Your Own Data

Every module traced back to something this cohort actually said, before a single slide existed.

- 🗨️

"18 of 19 declined a PR that merely passed tests"

→

✅

Module 09's evidence-based quality gate
- 🗨️

"Learn MCP and agentic workflows" — the #1 ask, 13 of 19

→

✅

Module 07's governed MCP workflows
- 🗨️

"15 of 19 already work from some form of spec"

→

✅

Module 04's Spec-Driven Development
- 🗨️

"7 of 19 called their testing 'mostly manual'"

→

✅

Module 06's automated test strategy
- 🗨️

"AI observability ranked dead last — 2 of 19"

→

✅

Module 10's Agent Prism, built anyway
- 🗨️

"Reduce feature delivery time by 30–50%"

→

✅

Module 11's ROI model target

The point isn't the coincidence: it's that good engineering judgment was already in the room. This programme's job was to formalize it, not import it.

What the Capstone Deliverable Contains

Ten components, one integrated use case — every module leaves a fingerprint.



Copilot Workflow

Module 02



Embedded SDD

Module 04



Greenfield or Brownfield Approach

Module 05



Software-Level Test Strategy

Module 06



Reusable Embedded Skill Pack

Module 07



MCP-Enabled Workflow

Module 07



PR Quality Automation

Module 09



LLM-as-Judge Evidence

Module 09



Context / Token Discipline

Modules 03, 11



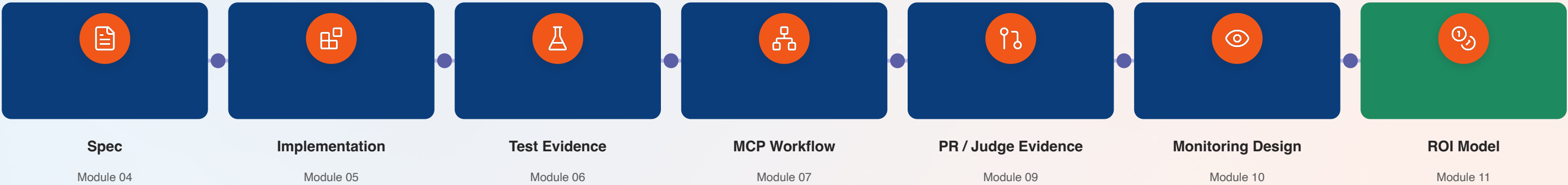
Agent Prism Monitoring Design

Module 10

Plus two things numbers alone don't capture: a business case (Module 11) and a governed enterprise rollout roadmap — covered next.

The Solution Blueprint: One Use Case, Fully Traced

The traceability chain from Module 04, extended all the way to a business decision.



The test your blueprint has to pass: can a reviewer follow this exact chain from your spec to your ROI number without a gap? If yes, the capstone is doing its job.

Worked Example: Where One Requirement Ends Up

REQ-014 → implemented in can_driver.c → TC-014a proves it → the MCP workflow that built it left a trace → the PR carries Judge evidence → Agent Prism shows its token cost → that cost feeds the cost-per-accepted-output line in your ROI model.

Choosing Your Use Case: Greenfield or Brownfield

Module 05's distinction, one more time — pick the one your team can trace most convincingly.

PICK GREENFIELD IF...



You have a clean, well-scoped spec from Module 04 ready to build from.



Your team wants to show the full SDD → implementation → test arc with nothing inherited.



The strongest story is speed and cleanliness, start to finish.

PICK BROWNFIELD IF...



You have a real repository archaeology story — architecture discovered, not designed fresh.



Your team wants to prove regression discipline: full suite passed, behaviour preserved.



The strongest story is judgment under constraint — exactly what leadership worries about most.

Evaluation Rubric: What “Excellent” Looks Like

Five dimensions, scored the way Module 09 taught you to score anything — with evidence.



Traceability

Every claim traces to an artifact — spec, test, PR, or trace — nothing asserted without evidence.



Technical Soundness

The implementation is correct, tested, and respects the minimal-change discipline where relevant.



Governance Discipline

Permissions, approvals & escalation paths are named, not assumed — Module 07's boundaries held.



Business Case Clarity

Cost per accepted output and effort saved are calculated, not estimated by feel.



Roadmap Realism

The 90-day plan names owners, milestones & risks — not just “scale it up.”

The 90-Day Adoption Roadmap

Three stages, growing scope — the plan every capstone hands to a technical owner.



Days 1–30

Prove It Again, Deliberately

- Re-run the capstone workflow on a second, independent use case
- Confirm the ROI numbers hold outside the demo conditions
- Name the technical owner for ongoing maintenance



Days 31–60

Extend to the Team

- Onboard 2–3 more engineers onto the same MCP workflow & skill pack
- Agent Prism thresholds tuned from real, not simulated, usage
- First governance review against Module 11's report format



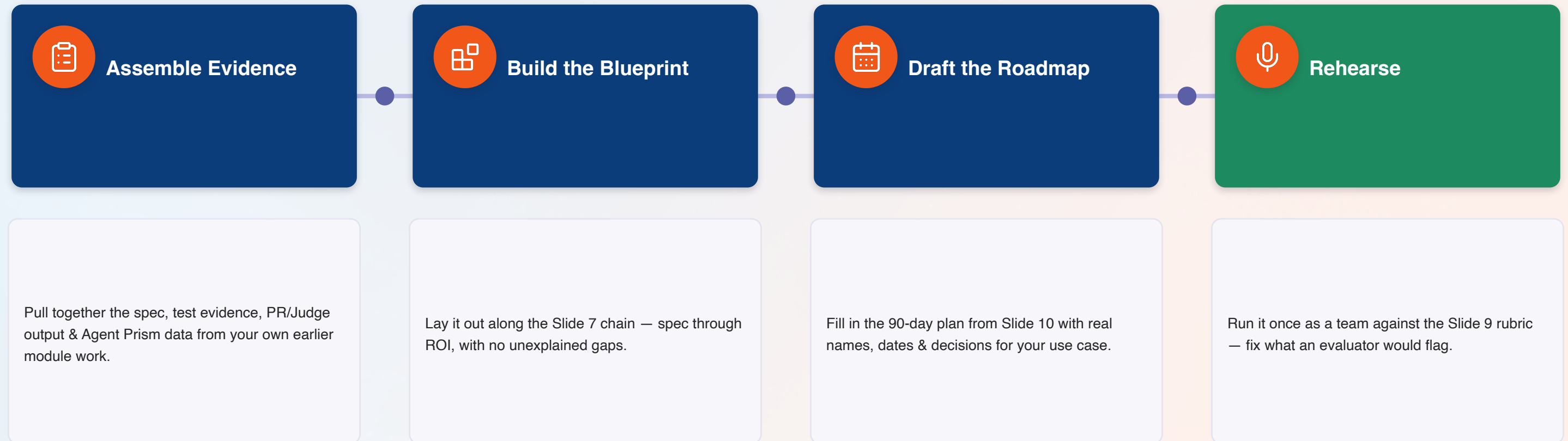
Days 61–90

Extend Across the GBE

- Roll out to a second team within the same Business Unit
- Formal scale / tune / restrict / retire decision, evidence-backed
- Handoff package ready for the next GBE to adopt

Hands-On: Building Your Capstone Presentation

Four stages, sixty minutes — from scattered artifacts to a story you can defend.



Facilitators circulate throughout: use them — a gap caught now is a question you won't get asked cold in front of the room.

Presentation Format & Time Allocation

What to expect when it's your team's turn — no surprises.



8 Minutes to Present

Walk the blueprint chain from spec to ROI — evaluators follow along on the rubric.



4 Minutes of Questions

Expect at least one traceability question — “show me the evidence for that claim.”



Scored Live, Against Slide 9

Every dimension scored in the room — no delayed or private evaluation.



Feedback Before You Sit Down

One strength and one gap, named out loud — useful even after the programme ends.

This isn't a pass/fail gate: it's the same evidence-based review discipline from Module 09, aimed at your own work for once.

Anti-Patterns: How Capstones Fall Short

Five habits that turn a strong programme's worth of work into a weak final impression.

DO THIS

-  Trace every claim to a real artifact from your own module work
-  Pick the use case with the strongest evidence, not the flashiest
-  Show the ROI math, including where the numbers came from
-  Name real owners & dates in the 90-day roadmap
-  Treat a rubric gap as useful feedback, not a personal critique

NOT THAT

-  Describe what the workflow “would” do, without showing it
-  Choose scope you can't actually back up in 8 minutes
-  State a percentage improvement with no visible calculation
-  Write “scale across the organization” with no next concrete step
-  Get defensive about a traceability question instead of answering it

Programme Outcomes: What This Cohort Built

Twelve modules, twenty-eight engineers, one AI-assisted embedded engineering practice — proven, not promised.

- ★ A shared vocabulary for AI Champion, Evangelist & Sponsor roles across PA, IA & BA
- ★ Fluency with GitHub Copilot Enterprise's full capability map, calibrated to embedded reality
- ★ A working spec-driven development practice with GitHub Spec Kit / OpenSpec
- ★ Proven greenfield build and brownfield safe-evolution workflows
- ★ An embedded test strategy that reliably catches a planted defect
- ★ Governed MCP workflows connecting agents to real engineering tools
- ★ An automated, evidence-based PR quality gate with LLM-as-Judge
- ★ Full observability into agent activity, cost & adoption via Agent Prism
- ★ A real ROI model and a scale/tune/restrict/retire governance habit
- ★ A complete, presented capstone — proof the whole practice holds together



30-DAY CATCH-UP REVIEW & COMPLIMENTARY MASTERCLASS STILL AHEAD — SEE MODULE 01



Congratulations

You've completed the Honeywell AI Champions Programme

Twelve modules. Twenty-eight engineers across PA, IA & BA. One AI-assisted embedded engineering practice — built on your own calibration data, proven in your own capstones.